

SE-101 Engineering Graphics and Design

Fall 2023: Team BBE-2

Interim Report 1

Multifunctional Food Processor



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Introduction of the Product:

This product is a multifunctional food processor. As the name suggests, it can perform various functions, like dehydrating vegetables, separating egg yolk from egg white, and blending food.

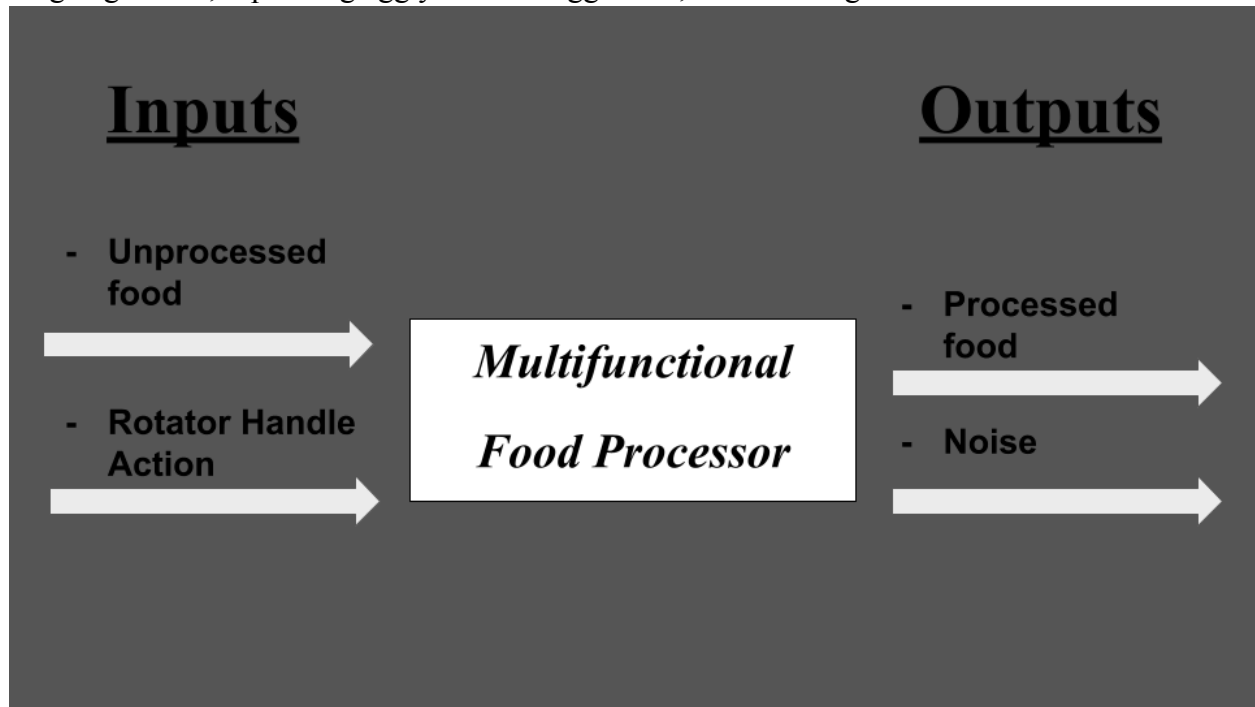


Figure 1. Black Box Diagram of the Multifunctional Food Processor

Target Buyer:

Home cooks are the target buyers. It is common to see different utensils used for even the most trivial of tasks, thus causing congestion at the kitchen counter. Therefore, to enhance customer convenience, this product performs many tasks with simple tool changes within the food processor.

One advantage this product has over most other products (like the one shown below) is that it is hand powered. So, the customer does not depend on the availability of electricity for using this product. Moreover, it is of a reasonable size and is thus portable.



(comparable product)



(our product)

Though the product is less robust than the one shown in the picture, it would be more cost effective since it provides more functionalities at a lower cost.

Dissection Process:

Initial Dissection:

Step 0: Initial Product.



Step 1: Separate the head part and base part from each other. Put aside the garlic peeler and the cleaning brush.

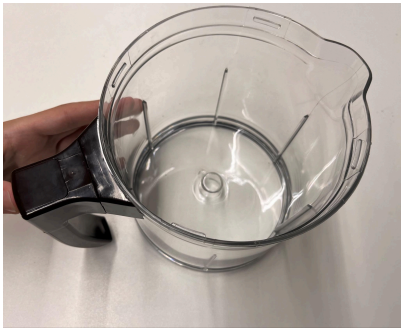


Figure 2. Base Part



Figure 3. Head Part



Figure 4. Garlic Peeler



Figure 5. Cleaning Brush

Head Part Dissection:

Step 1: Separate the head plate, egg separator, and rotator handle. Put aside the egg separator.



Figure 6. Head Plate



Figure 7. Egg Separator



Figure 8. Rotator Handle

Step 1.1: Unscrew the screw connecting the Rotator Base and the Handle Bulb.



Figure 9. Rotator Base



Figure 10. Handle Bulb

Step 2: Unscrew the 7 screws that connect the Head Plate Top and the Head Plate Bottom. Put aside Head Plate Bottom.



Figure 11. Head Plate Bottom



Figure 12. Head Plate Top (inside)

Step 3: Separate the Head Plate Top and the 3 Gears (Slow, Medium, Quick). Put aside all.



Figure 13. Head Plate Top without gears



Figure 14. Slow Gear



Figure 15. Medium Gear

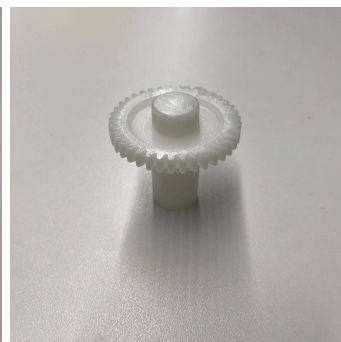


Figure 16. Quick Gear

Base Part Dissection:

Step 1: Separate the Clear Container (with the handle), Egg Blender, Blade Unit, Dehydrator, Extender. Set aside Egg Blender, Dehydrator, Extender.

*Note: The Extender piece is used for both the Blade Unit and the Dehydrator



Figure 17. Clear Container (with retractable handle)



Figure 18. Egg Blender



Figure 19. Blade Unit (top view)



Figure 20. Dehydrator



Figure 21. Extender



Figure 22. Blade Unit (bottom view)

Step 2: Unscrew the screw connecting the blade holder and each blade. Set the blade holder and the blades aside.



Figure 23: Blade Holder



Figure 24: Blade Holder with blades (4)



Figure 25: Blade

Step 3: Separate the Retractable Handle (Screw Mount connected to Handle Base) and the Clear Pitcher. When separating, the Handle Connector will be connected to the Clear Pitcher. Unscrew the 4 screws connecting both parts. Set aside handle connector and clear pitcher.



Figure 26: Clear Pitcher with Handle Connector



Figure 27: Handle Connector



Figure 27: Clear Pitcher (left) and Handle Connector (right)



Figure 28: Retractable Handle with Handle Connector

Step 4: Unscrew the screw connecting the Handle Base and the Screw Mount. Set aside Handle Base and Screw Mount.



Figure 29: *Handle Base (left)
Screw Mount (right)*

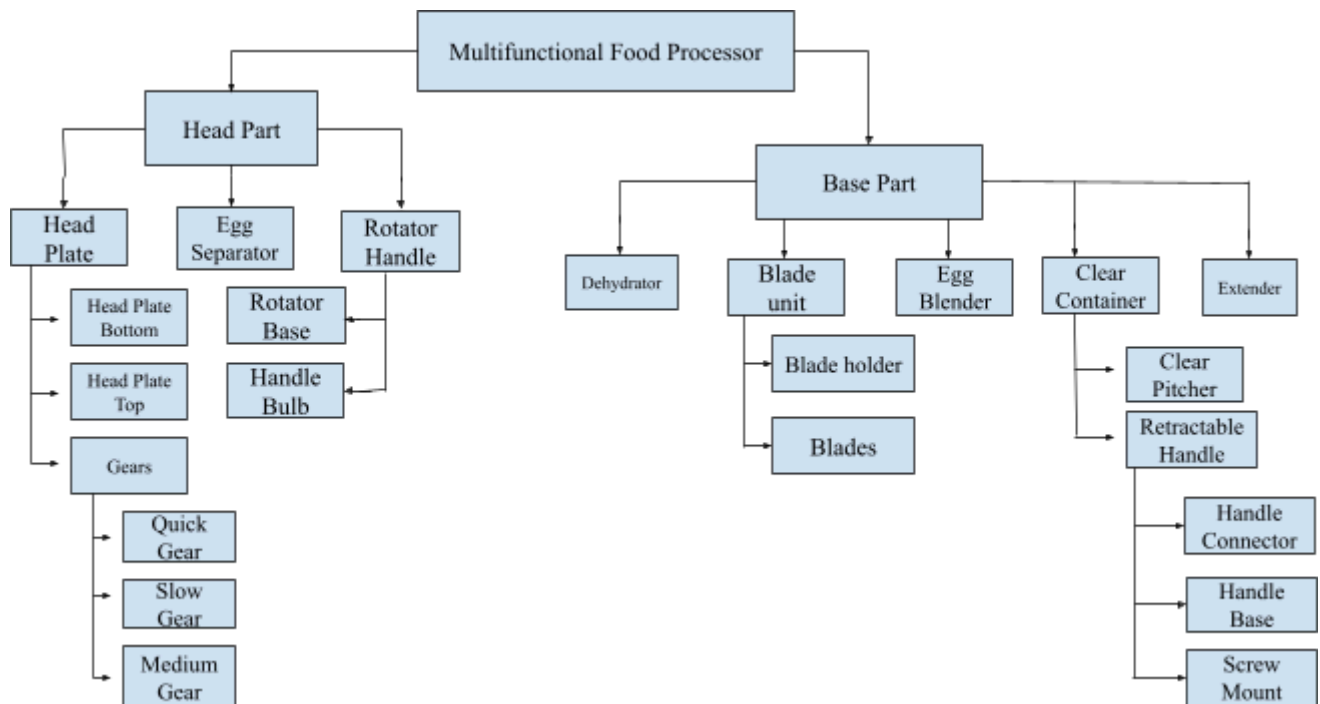


Figure 30: *Product Decomposition Diagram*

References

Bezos, Jeff. "Hamilton Beach Food Processor & Vegetable Chopper For ..." *Amazon*,
Hamilton Beach, 9 Nov. 2012,
www.amazon.com/Hamilton-Beach-70730-Processor-Vegetable/dp/B008J8MJIQ.